

Eros Fani



📅 July 13, 1997 ✉ eros.fani@gmail.com 🌐 erosfani.github.io in LinkedIn 🐙 Github

Professional and Teaching Experience

Machine Learning Researcher

GENSYN

Oct 2025

→ Jul 2026

- Led research on **model merging**, **mixture of experts**, **LLMs routing**, and **decentralized ML**
- Designed and implemented a **decentralized AI inference platform** that orchestrates multiple remote LLM providers behind a unified conversational interface (public release forthcoming)
- Wrote technical blog posts [🔗](#) [🔗](#)
- Invited as a speaker at two Gensyn X Space events [🔗](#) [🔗](#), each later featured in a Gensyn article [🔗](#) [🔗](#)

Postdoctoral Fellow & Software Engineer

BASQUE CENTER FOR APPLIED MATHEMATICS (BCAM)

Nov 2024

→ Oct 2025

- Lead developer, **Act.AI project** (PI: Prof. N. Quadrianto), on the development of a **unified framework** for LLMs for **tabular data**

Senior Researcher

PREDICTIVE ANALYTICS LAB, UNIVERSITY OF SUSSEX (UoS)

May 2024

→ Oct 2025

- Led research on mitigating the effects of **spurious correlations** in **federated learning**

MSc Thesis Supervisor

POLYTECHNIC UNIVERSITY OF TURIN (POLITO)

Sep 2023

→ Oct 2024

- Topic: Clustered Federated Learning • Resulted in a publication at **ICML25**

Teaching Assistant

POLYTECHNIC UNIVERSITY OF TURIN (POLITO)

Sep 2022

→ Sep 2023

- Graduate courses: Machine Learning and Deep Learning ◦ Advanced Machine Learning ◦ Data Analytics and Artificial Intelligence ◦ Web Applications I • Undergraduate course: Computer Science

MSc Thesis Supervisor

POLYTECHNIC UNIVERSITY OF TURIN (POLITO)

Sep 2022

→ Jul 2023

- Topic: Federated Visual Geo-Localization • Resulted in a publication at **FedVision @ CVPR 2024**

Teaching Collaborator as an Undergrad

POLYTECHNIC UNIVERSITY OF TURIN (POLITO)

Sep 2018

→ Mar 2019

- Undergraduate courses: Algorithms and Programming ◦ Mathematical Analysis I ◦ Mathematical Analysis II

Education

Ph.D. in Computer and Control Engineering

VISUAL AND MULTIMODAL APPLIED LEARNING LABORATORY (VANDAL), POLYTECHNIC UNIVERSITY OF TURIN (POLITO)

Nov 2021

→ Jul 2025

- Thesis title: **Addressing Heterogeneity in Federated Learning for Real-world Vision Applications**
- Advisor: Prof. B. Caputo · Polytechnic University of Turin (PoliTo)

ELLIS Ph.D. Program - Academic Track

EUROPEAN LABORATORY FOR LEARNING AND INTELLIGENT SYSTEMS (ELLIS)

Nov 2021

→ Jul 2025

- Primary Host: Prof. B. Caputo · Polytechnic University of Turin (PoliTo)
- Exchange Host: Prof. N. Quadrianto · University of Sussex (UoS)

Visiting Period at the University of Sussex for the ELLIS Exchange

UNIVERSITY OF SUSSEX (UoS), EUROPEAN LABORATORY FOR LEARNING AND INTELLIGENT SYSTEMS (ELLIS)

May 2024

→ Oct 2024

- Research Advisor: Prof. N. Quadrianto · University of Sussex (UoS)

Visiting Period at ELLIS Alicante

ELLIS ALICANTE, EUROPEAN LABORATORY FOR LEARNING AND INTELLIGENT SYSTEMS (ELLIS)

Sep 2024

- Research Advisor: Prof. N. Oliver · ELLIS Alicante

MSc in Data Science and Engineering POLYTECHNIC UNIVERSITY OF TURIN (POLiTo) - Final grade: 110/110 cum laude - Thesis title: On the Challenges of Class Imbalance in Federated Learning for Semantic Segmentation	Oct 2019 → Oct 2021
BSc in Computer Engineering POLYTECHNIC UNIVERSITY OF TURIN (POLiTo) Final grade: 108/110	Oct 2016 → Jul 2019
Young Talent Project POLYTECHNIC UNIVERSITY OF TURIN (POLiTo), CRT FOUNDATION Study program of excellence open only to the 2% top-tier students admitted to the PoliTo for the A.Y. 2016/17	Oct 2016 → Jul 2019
High School Diploma LICEO SCIENTIFICO STATALE M. CURIE GIULIANA Final grade: 100/100 cum laude	Sep 2011 → Jul 2016

Publications

E. Fani , O. Ersoy. “ <i>IR3DE: A Linear Router for Large Language Models</i> ”. Resource-Adaptive Foundation Model Inference (AdaptFM) workshop of the International Conference on Machine Learning, 2026.	AdaptFM ICML26
E. Fani , O. Ersoy. “ <i>Training-Free Dynamic Upcycling of Expert Language Models</i> ”. Scaling Post-training for LLMs (SPOT) workshop of the International Conference on Learning Representations, 2026.	SPOT ICLR26
Extended version under review at a major ML conference.	
G. D. Németh, E. Fani , et. al. “ <i>Diversity-Driven Learning: Tackling Spurious Correlations and Data Heterogeneity in Federated Models</i> ”. Third IEEE International Conference on Federated Learning Technologies and Applications, 2025.	FLTA25
E. Fani , et. al. “ <i>Resource-Efficient Personalization in Federated Learning With Closed-Form Classifiers</i> ”. IEEE Access, 2025	IEEE ACCESS
A. Licciardi, ..., E. Fani , et. al. “ <i>Interaction-Aware Gaussian Weighting for Clustered Federated Learning</i> ”. Forty-second International Conference on Machine Learning, 2025.	ICML25
E. Fani , et. al. “ <i>Accelerating Heterogeneous Federated Learning with Closed-form Classifiers</i> ”. Forty-first International Conference on Machine Learning, 2024.	ICML24
M. Dutto, ..., E. Fani , et. al. “ <i>Collaborative Visual Place Recognition through Federated Learning</i> ”. FedVision Workshop in Conjunction with the IEEE/CVF Conference on Computer Vision and Pattern Recognition, 2024.	FEDVISION CVPR24
E. Fani , et. al. “ <i>Fed3R: Recursive Ridge Regression for Federated Learning with strong pre-trained models</i> ”. International Workshop on Federated Learning in the Age of Foundation Models in Conjunction with NeurIPS, 2023.	FL@FM NEURIPS23
E. Fani , et. al. “ <i>FedDrive v2: an Analysis of the Impact of Label Skewness in Federated Semantic Segmentation for Autonomous Driving</i> ”. 5th Italian Conference on Robotics and Intelligent Machines, 2023.	I-RIM23
D. Shenaj*, E. Fani *, et. al. “ <i>Learning Across Domains and Devices: Style-Driven Source-Free Domain Adaptation in Clustered Federated Learning</i> ”. IEEE/CVF winter conference on applications of computer vision, 2023.	WACV23
L. Fantauzzo*, E. Fani *, et. al. “ <i>FedDrive: Generalizing Federated Learning to Semantic Segmentation in Autonomous Driving</i> ”. IEEE/RSJ International Conference on Intelligent Robots and Systems, 2022.	IROS22

*Equal contribution.

Selected Honors and Awards

Invited to Present at the Flower AI Summit, Organized by Flower AI 🔗 Presenting the Research Paper: TRAINING-FREE DYNAMIC UPCYCLING OF EXPERT LANGUAGE MODELS	Apr 2026
Outstanding Reviewer award at ECCV24 EUROPEAN CONFERENCE ON COMPUTER VISION 2024, EUROPEAN COMPUTER VISION ASSOCIATION (ECVA)	Sep 2024

Languages

Italian - native

English - full professional proficiency

Spanish - elementary proficiency

Coding Skills

Python, C, PyTorch, Numpy, Scikit-Learn, WandB, Matplotlib, Kubernetes, HTML, HTML5, HTML Bootstrap, CSS, Jax, FedJax, Haiku, Flax, Flower, Textual, Java, C++, Bash, Matlab, Javascript, ReactJS, Vite, PHP, R, Apache Spark, Apache Hadoop